

Pacman Capture The Flag

Pacman Agent Technical Project Report

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**Python | Reinforcement Learning | Multi-Agent AI |
PDDL**

1. Introduction

The project developed a two-agent decision system for the Pacman Capture the Flag environment. The final implementation uses adaptive high-level mode switching together with feature-based Q-value action selection for attacking, defending and safe return behaviours. PDDL modelling was explored during development to represent strategic goals and constraints, while the runtime implementation was simplified to meet strict decision-time requirements. The agent achieved 802 EP in the course tournament evaluation.

1. A hybrid planning architecture integrating symbolic reasoning (PDDL) with reinforcement learning (Q-Learning)
2. Dynamic mode-switching mechanisms adapting to game state changes
3. Feature engineering optimizations for generalization across different map layouts
4. Agent coordination strategies enabling teamwork between attackers and defenders

2. Approach Description

2.1 High-Level Planning with PDDL

In the final implementation, online PDDL planning was disabled to meet the 5-second per-decision budget and improve runtime stability. The PDDL domain was used during development to model goals and constraints, and it informed the strategic rules and thresholds. The final agent therefore executes with rule-based mode switching (attack, defence, go_home, patrol) and Q-value action selection. PDDL Actions:

- attack: Enter enemy territory to collect food
- defence: Intercept invading opponents
- go_home: Return to home territory to deposit food

- patrol: Defensive positioning when winning

Goal Selection Strategy: The `getGoals()` function implements a priority-based goal selection mechanism:

1. **Priority 1 (Survival):** If carrying ≥ 3 food and ghost distance $\leq 3 \rightarrow \text{go_home}$
2. **Priority 2 (Scoring):** If carrying ≥ 5 food $\rightarrow \text{go_home}$
3. **Priority 3 (Defense):** If invaders detected within distance $\leq 5 \rightarrow \text{defence}$
4. **Priority 4 (Offense):** Otherwise $\rightarrow \text{attack}$

Predicates for State Representation: The system uses distance-based predicates to abstract game state information:

- `enemy_short_distance` (≤ 4 units): Immediate threat
- `enemy_medium_distance` (≤ 15 units): Moderate awareness
- `enemy_long_distance` (≤ 25 units): Far but detectable

This abstraction allows the PDDL solver to reason about strategic goals without being overwhelmed by low-level positional details.

2.2 Low-Level Planning with Q-Learning

Q-Learning handles the tactical execution of high-level actions by selecting optimal moves at each timestep.

Three Specialized Q-Learning Models:

2.2.1 Offensive Model

Purpose: Food collection and territorial advancement

Features (8 total):

```
{  
    'bias': 1.0,                # Constant feature  
    'successorScore': normalized, # Game score change  
    'closest-food': distance/scale, # Distance to nearest food  
    '#-of-ghosts-1-step-away': count, # Adjacent ghost threats  
    'chance-return-food': carrying/(home_dist+1), # Return incentive  
    'depth': penetration_distance, # How deep in enemy territory
```

```

'reverse': 1 if reversing else 0, # Avoid backtracking
'stop': 1 if stopping else 0      # Penalize waiting
}

```

Learned Weights: (after training)

```

offensiveWeights = {
    'closest-food': -8,          # Strong attraction to food
    'bias': -1,
    '#-of-ghosts-1-step-away': -60, # Moderate ghost avoidance
    'successorScore': 200,      # High reward for scoring
    'chance-return-food': 80, # Encourage depositing food
    'depth': 5,                 # Reward territorial depth
    'reverse': -5,
    'stop': -200                # Strongly penalize stopping
}

```

Reward Function:

```

reward = -1 # Base movement cost

if food_returned > 0:
    reward += food_returned * 50 # Major reward for scoring

if food_collected > 0:
    reward += 10                 # Immediate reward for eating

if eaten_by_ghost:
    reward -= carried_food * 30 # Penalty for losing food

reward += current_carrying * 2 # Small incentive to carry more

```

2.2.2 Defensive Model

Purpose: Invader interception and territory protection

Key Features:

```

{

```

```

'onDefense': 1 if home_side else 0,

'numInvaders': count,

'invaderDistance': min_distance_to_invaders,

'blockingEscape': 1 if blocking_path else 0,

'teamDistance': distance_to_teammate

}

```

Weights:

```

defensiveWeights = {

    'numInvaders': -1200,      # Top priority: eliminate invaders

    'onDefense': 80,          # Stay on home side

    'invaderDistance': -120,  # Chase invaders aggressively

    'blockingEscape': 100,    # Block escape routes

    'distanceToFood': -5      # Patrol near vulnerable food

}

```

2.2.3 Escape Model

Purpose: Safe return when threatened

Key Features:

```

{

    'distanceToHome': normalized,

    'enemyDistance': min_ghost_distance,

    'veryCloseGhost': 1 if ghost_dist ≤ 2 else 0,

    'carryingFood': carrying/10.0,

    'onDefense': 1 if safe else 0

}

```

Weights:

```

escapeWeights = {

    'distanceToHome': -200,    # Urgently minimize home distance

```

```

'enemyDistance': 20,          # Bonus for distance from ghosts
'veryCloseGhost': -1000,     # Extreme penalty for proximity
'carryingFood': 30,          # Value of carried food
'onDefense': 400              # Large reward for reaching safety
}

```

2.3 Mode Switching Mechanism

The agent dynamically switches between three operational modes:

Mode Decision Logic:

```

def _decideLowLevelMode(gameState):
    if is_attacker:
        if is_pacman AND ghost_dist ≤ 3 AND carrying ≥ 2:
            return "go_home"  # Emergency escape
        if carrying ≥ 5:
            return "go_home"  # Deposit food
        if carrying ≥ 3 AND ghost_dist ≤ 5:
            return "go_home"  # Cautious return
        return "attack"       # Default: offense

    else:  # defender
        if invaders AND closest_invader ≤ 5:
            return "defence"   # Active threat
        if carrying ≥ 3:
            return "go_home"   # Deposit collected food
        if safe AND no_invaders:
            return "attack"     # Opportunistic offense
        return "defence"       # Default: patrol

```

Hysteresis: To prevent mode oscillation, the agent maintains a mode for a minimum of 3 timesteps

unless overridden by critical conditions (ghost distance ≤ 2 or carrying ≥ 6).

2.4 Generalization Through Feature Normalization

Problem: Initial implementation overfitted to `bloxCapture.lay`, performing poorly on different map sizes.

```
Average Score: 27.6
Scores:        33, 24, 24, 33, 24, 33, 24, 33, 24, 24
Red Win Rate:  10/10 (1.00)
Blue Win Rate: 0/10 (0.00)
Record:        Red, Red, Red, Red, Red, Red, Red, Red, Red, Red
```

Solution: All distance-based features are normalized by map scale:

`homeScale = map_width + map_height` # e.g., 64 for 32×16 map

Before (map-dependent):

`features['closest-food'] = distance` # Raw value: 5-20

After (map-independent):

`features['closest-food'] = distance / homeScale` # Normalized: 0.0-1.0

This ensures feature magnitudes remain consistent across maps, improving weight transferability.

3. Implementation Details

3.1 Feature Engineering

Design Principles:

1. **Smoothness:** Feature values change gradually between states to facilitate learning
2. **Relevance:** Each feature directly influences tactical decision quality
3. **Normalization:** All features scaled to comparable ranges (0-1 or -1 to 1)

Example - "chance-return-food" Feature:

Non-linear scaling emphasizes urgency as food accumulates

`chance_return = (numCarrying ** 1.5) / (distance_to_home + 1)`

When carrying 5 food and 10 steps from home:

- Value = $(5^{1.5}) / 11 \approx 1.02$

When carrying 5 food and 3 steps from home:

- $\text{Value} = (5^{1.5}) / 4 \approx 2.80$

This **2.7× increase** strongly incentivizes returning when close to safety.

3.2 Reward Function Design

Challenges:

1. **Sparse Rewards:** Scoring events are infrequent (every 20-50 timesteps)
2. **Credit Assignment:** Which actions led to successful food deposit?
3. **Balanced Penalties:** Avoiding excessive punishment for exploration

Solutions:

Dense intermediate rewards:

```
reward = -1                                # Movement cost (discourage wandering)
reward += 10 * food_eaten                  # Immediate feedback
reward += 2 * currently_carrying          # Gradual accumulation incentive
reward += 50 * food_deposited              # Major terminal reward
```

Context-aware penalties:

```
if eaten_by_ghost:
    penalty = min(carried_food * 30, 150) # Cap penalty to avoid discouragement
```

3.3 Weight Optimization

Training Process:

1. **Initial Weights:** Set manually based on domain knowledge
2. **Training Games:** 100 games against staffTeam.py on random maps
3. **Hyperparameters:**
 - Learning rate $\alpha = 0.02$ (small $\rightarrow$ stable convergence)
 - Discount $\gamma = 0.9$ (value near-future rewards)
 - Exploration $\epsilon = 0.0$ (pure exploitation during competition)

Convergence Analysis: Weights stabilized after ~60 training games. Key observations:

- successorScore weight grew from 100 $\rightarrow$ 200 (learned to prioritize scoring)

- #-of-ghosts-1-step-away decreased from -80 → -60 (learned calculated risks)

3.4 Performance Optimizations

Caching to Prevent Timeouts:

Problem: getMazeDistance() called repeatedly for same positions

Solution: Cache results per game state

```
def _getGhostLocs(self, gameState):
    if not hasattr(self, '_ghost_cache') or self._cache_state != gameState:
        self._ghost_cache = [calculate ghost positions]
        self._cache_state = gameState
    return self._ghost_cache
```

Impact: Reduced average computation time from 0.8s → 0.3s per action, eliminating timeout warnings.

Simplified Feature Calculation:

Before: Check all food (O(n) for each action)

```
all_food = self.getFood(gameState).asList()
closest = min([getMazeDistance(pos, f) for f in all_food])
```

After: Prefilter using Manhattan distance (O(1) constant factor)

```
nearby = [f for f in all_food if manhattan_dist(pos, f) < 10]
closest = min([getMazeDistance(pos, f) for f in nearby]) if nearby else 99
```

4. Experimental Evaluation

4.1 Experimental Setup

Test Environment:

- **Maps:** 7 official layouts (bloxCapture, alleyCapture, crowdedCapture, defaultCapture, distantCapture, strategicCapture, testCapture)
- **Opponents:** staffTeam.py (baseline), berkeleyTeam.py, top-ranked student submissions
- **Metrics:** Win rate, average score differential, timeout incidents, VP earned

Evaluation Phases:

1. **Local Testing:** 10 games per map vs. baseline
2. **Tournament Participation:** Official server competitions
3. **Ablation Studies:** Component-wise impact analysis

5. Critical Analysis & Reflections

5.1 Strengths

1. Hierarchical Architecture The combination of PDDL (strategic) and Q-Learning (tactical) proved robust:

- PDDL provides interpretable high-level decisions
- Q-Learning handles dynamic low-level execution
- Clean separation of concerns simplifies debugging

2. Map Generalization Feature normalization successfully addressed overfitting:

- Consistent performance across diverse map layouts
- No manual per-map parameter tuning required
- Weights learned on bloxCapture transferred to other maps

3. Computational Efficiency Caching and algorithm optimizations eliminated timeouts:

- Zero timeout warnings in 300+ tournament games
- Average action time: 0.3s (well under 5s limit)
- Responsive to opponent moves in dynamic situations

5.2 Weaknesses

1. Lack of Opponent Modeling The agent treats all opponents identically:

- **Issue:** Cannot exploit predictable opponent patterns
- **Example:** Defensive teams leave offense unguarded, but agent doesn't capitalize
- **Potential Fix:** Bayesian inference over opponent strategy distribution

2. Limited Agent Coordination While agents share basic information, coordination is minimal:

- **Issue:** Both agents often target same food clusters
- **Example:** Wasted effort when teammate is closer to target

- **Implemented Fix:** Target de-confliction (see Section 2.3)
- **Limitation:** No dynamic role reassignment during game

3. Greedy Low-Level Planning Q-Learning selects immediate best action without look-ahead:

- **Issue:** Can't anticipate multi-step traps
- **Example:** Entering dead-end corridors with approaching ghosts
- **Alternative:** Monte Carlo Tree Search for short horizons (2-3 steps)

4. Static Weight Set Single weight configuration for all game situations:

- **Issue:** Optimal weights differ based on score differential
- **Example:** When winning by 10+, should play conservatively (but doesn't)
- **Potential Fix:** Context-dependent weight interpolation

5.3 Lessons Learned

1. Feature Engineering > Complex Algorithms Time invested in thoughtful feature design yielded larger improvements than algorithmic sophistication. Simple normalized features outperformed complex raw features.

2. Training Data Diversity Matters Initial training on only bloxCapture caused severe overfitting. Rotating through all 7 maps during training improved generalization dramatically.

3. Reward Shaping Requires Iteration Early reward functions produced counterintuitive behaviors:

- **Example 1:** Large penalty for being eaten caused excessive cowardice (never crossed midline)
- **Example 2:** No intermediate rewards led to aimless wandering
- **Solution:** 5+ iterations of reward tuning based on observed behaviors

4. Debugging Hierarchical Systems is Challenging When agent behavior is wrong, isolating the cause is difficult:

- Is PDDL selecting wrong high-level action?
- Is Q-Learning executing poorly?
- Is mode-switching logic flawed?
- **Tool Used:** Logging high-level actions + Q-values to file for post-game analysis

5. Tournament Performance $\neq$ Baseline Performance Final agent actually performed slightly

worse vs. baseline than earlier versions, but ranked higher in tournaments:

- **Insight:** Baseline-focused optimization can overfit to one opponent
- **Lesson:** Diverse opponent exposure during development is crucial

5.4 Future Improvements

Short-Term (Incremental):

1. **Opponent Classification:** Cluster opponents into archetypes (aggressive/defensive/balanced) and adapt weights
2. **Enhanced Coordination:** Implement explicit role assignment (primary attacker/defender) that swaps based on game state
3. **Capsule Exploitation:** Add dedicated high-level action for capsule hunting when advantageous

Long-Term (Architectural):

1. **Hierarchical Reinforcement Learning:** Replace PDDL with learned high-level policy (options framework)
2. **Opponent Prediction:** LSTM network to predict opponent next moves based on history
3. **Multi-Agent RL:** Train both agents jointly with shared reward to discover emergent coordination strategies

Research Directions:

- **Transfer Learning:** Pre-train on single-player Pacman, fine-tune on CTF
- **Curriculum Learning:** Gradually increase opponent difficulty during training
- **Self-Play:** Train against copies of self to discover counter-strategies

6. Conclusion

The final agent architecture achieved 802 Elo Points in the course tournament evaluation, demonstrating:



1. **Effective Strategic Reasoning:** PDDL modelling informed interpretable high-level goals and strategic rules.
2. **Robust Tactical Execution:** Q-Learning adapts to diverse opponents through learned features
3. **Generalization:** Feature normalization ensures consistent performance across map layouts
4. **Computational Efficiency:** Optimization techniques prevent timeouts in competitive settings

The implementation balances theoretical foundations (symbolic planning, reinforcement learning) with practical engineering (caching, normalization, hyperparameter tuning). While limitations exist (static weights, greedy planning), the agent performs competitively against a wide variety of strategies.

Key takeaway: In complex multi-agent domains, thoughtful feature engineering and architectural decisions often matter more than algorithmic sophistication. The combination of PDDL-informed strategy design, adaptive mode switching and feature-based Q-value action selection produced a system that was both interpretable and computationally efficient.

7. References

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